Classical Mechanics: Newtonian, Lagrangian, and Hamiltonian formulations

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Abstract

These notes serve to summarize, compare, and make more mathematically precise (with respect to most mechanics textbooks) the three main formulations of classical mechanics: Newtonian, Lagrangian, and Hamiltonian. There is also the Hamilton-Jacobi formulation, which is not talked about much in undergraduate mechanics courses but is actually very important to understand when one wants to understand Schrodinger quantum mechanics and the semiclassical analysis of path integrals at deeper levels. I plan to update these notes with the Hamilton-Jacobi formulation. The sympletic structure and mathematical formulation of Hamiltonian mechanics, arguably the highlights of these notes, is discussed in [some chapter/section/appendix].

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1 Introduction

There is no shortage of classical mechanics textbooks, articles, and papers - e.g., the famous canonical textbook [3]. There is also a lot of literature on the mathematical aspects of classical mechanics [1], [5], emphasizing methods from differential geometry.

Classical mechanics is the study of time evolution of classical systems. Roughly speaking, a classical system is a macroscopic collection of objects, such as point particles and/or continuum objects which can have stress, strain, tension, etc. Technically speaking, a classical system is a system which evolves in time as governed by some classical equations of motion. There are 4 different, but equivalent, formulations of classical mechanics: Newtonian, Lagrangian, Hamiltonian, and Hamilton-Jacobi formulations. The goals of each formulation are the same: to predict the time evolution of a system given information about it such as the forces acting on it or the various energies of the system. Mathematically, this amounts to solving differential equations, called *equations of motion*, for the coordinates or parameters that describe the system, as functions of time. The three formulations achieve this goal in conceptually

and mathematically different ways, but all come to the same solution for the classical time evolution of a given system.

Practically, the choice of which formulation of mechanics to use to solve a classical mechanics problem often comes down to which formulation produces the fastest result.

Philosophically, the three different formulations are quite different and interesting in their own rights. The fact that they equivalently describe the classical time evolution of a given system is somewhat incredible and perhaps philosophically interesting.

Theoretically/mathematically, the choice of which formulation to use is an important question. The conceptual and mathematical differences in the formulations (mainly the Lagrangian and Hamiltonian formulations) are very significant in the theory of physics, especially in the context of quantum mechanics. Most if not all theories of physics are formulated in the Lagrangian and/or Hamiltonian formulations, or their quantum analogues.

Notation and conventions

Throughout these notes, we will explicitly only discuss classical physical systems consisting of non-relativistic point particles. However, the mathematical formulation extended to extended objects or field theories is not much more complex, and the underlying ideas discussed in these notes still hold in more general systems. More general systems are naturally richer, though, with mathematically and physically more sophisticated objects (c.f., stress-energy tensor T_{ab}, Maxwell's electromagnetic field tensor F_{ab}, and the general moment of inertia for rotational dynamics).

A point on a manifold, $p \in \mathcal{M}$, can be written in terms of local coordinate systems, $x^k : \mathcal{M} \to \mathbb{R}^n$, as an array of components $x \circ p = (x^1(p), x^2(p), ..., x^n(p))$. In these notes, and as is often done in physics, we simply write the point p. in a coordinate system x, as $p = (p^1, p^2, ..., p^n)$.

In particular, the coordinates of a path $\gamma : \mathbb{R} \to \mathcal{M}$ can be written as an array of numbers $\gamma^k(t) = (x^k \circ \gamma)(t)$. [Note that $\gamma^k(t)$ depends on the coordinate system x chosen to represent γ, but it won't be necessary to be so pedantic in these notes.] Even less pedantically, often in physics one write $x^k(t)$ as the coordinates of a particle trajectory. What is really meant is $x^k(t) \equiv \gamma^k(t)$.

2 Newtonian:

In the Newtonian formulation of mechanics, we consider the forces, as described by vectors, that affect a given system. A general system may consist of many point particles, but we shall only consider the case of a single point particle, since the following discussion of a single particle can be easily generalized to general systems by appropriately incorporating sums into the equations of motion and laws of motion. The fundamental postulates, called *Newton's laws of motion*, of this formulation of mechanics are

1. Objects in constant motion stay in constant motion unless subject to external forces,
2. The acceleration of an object, with mass m and described by position vector $\mathbf{r}(t)$, is governed by $m\mathbf{a} = \mathbf{F}$, where $\mathbf{a} = \frac{d^2}{dt^2}\mathbf{r}$ is the acceleration of the object and $\mathbf{F} = \sum_k \mathbf{F}_k$ is the sum of all forces acting on the object,
3. For every force on an object, there is an opposite and equal force.

These three laws of motion are the substance of the Newtonian formulation. Practically, the procedure of solving a mechanics problem using this formulation consists of: 1. identifying the total force $\mathbf{F} = \sum_k \mathbf{F}_k$ acting on the system. [Note: the individual force vectors $\mathbf{F}_k$ might be functions of position, velocity, or even time.] 2. construct the equation of motion $\mathbf{m}\frac{d^2}{dt^2}\mathbf{r}(t) = \mathbf{F}$. 3. solve Newton's equation of motion for $\mathbf{r}(t) = (x(t), y(t), z(t))$.

In principle, Newton's equation can be solved for the path of a classical particle $\mathbf{r}(t)$ given the forces acting on the system $\mathbf{F}_k(x, y, z, t)$. This can, however, be challenging analytically, especially if there many force vectors described

by exotic functions.

Analytical mechanics: Lagrangian and Hamiltonian formulations

Although all three formulations of mechanics are different, the Lagrangian and Hamiltonian formulations are, in some sense, "transforms" of each other and thus are in principle simply related. This transform can be given abstractly as an isomorphism $\mathcal{L} : TC \to T^*C$, from the tangent bundle of configuration space (the mathematical arena of Lagrangian mechanics) to the co-tangent bundle of configuration space called phase space (the mathematical arena of Hamiltonian mechanics). Furthermore, and in particular, they are based upon the concepts of the energy of a system whereas the Newtonian formulation is based on the concept of force on a system. Therefore, the Lagrangian and Hamiltonian formulations are more similar to each other than they are to Newtonian formulation, and are called the analytical formulations of mechanics since they deal with scalar quantities, i.e., energy, as opposed to force vectors. This sometimes has practical advantages but the main advantages are often conceptual and theoretical. For example, by specifying a single scalar function in the Lagrangian formulation of mechanics, the physics of the whole universe can be in principle described, while specifying all force vectors in the universe is completely not tractable, perhaps not even in principle. The Lagrangian formulation also gives simple ways of study properties of a system, such as symmetries.

3 Lagrangian

A classical system can be described by a minimal set of parameters which reflect its degrees of freedom. For example, the parameters describing a single particle of mass m in 3-dimensional space can simply be the cartesian coordinates of the particle $\mathbf{r}(t) = (x(t), y(t), z(t))$ or those of the spherical coordinate system, $\mathbf{r}(t) = (r(t), \theta(t), \phi(t))$. In general, if a system of particles has n degrees of freedom, then it can be described by n *generalized coordinates* $q = (q^1, q^2, ..., q^n)$. These n variables are the minimal number of variables needed to describe a configuration of a system, and are, in general, fewer in quantity than the $3N$ cartesian coordinates $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N)$ used to describe the positions of each particle in a system. The process of systematically reducing the $3N$ cartesian coordinates to the n generalized coordinates q^k is formally accomplished by solving constraints, which are equations of the form $F(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N, t) = 0$. If there are k such constraints, then there are $n = 3N - k$ generalized coordinates. That is, a system has $3N - k$ degrees of freedom in general. Practically, mathematically, and conceptually it is much easier/simpler/better to work with the minimum number of degrees of freedom describing the system than all of the cartesian coordinates of the particles making up the system.

Since particles in a system evolve in time along trajectories, or histories, the generalized coordinates which represent the time evolution of the system obviously also depend on time: $q(t) = (q^1(t), q^2(t), ..., q^n(t))$. Using these minimal degrees of freedom, we can define the *configuration* of a given system at some time t as a point $q(t)$ in the *configuration space* C of the system, which is the space of all possible values of the generalized coordinates q^k. That is, C is the space of all possible physical configurations of a system. [Oftentimes, C can be treated as flat Certesian space $\mathbb{R}^n$, but in general it can be a curved manifold, like a torus, $C = S^1 \times S^1$, as it is for the double pendulum. Nonetheless, since every manifold has an atlas, by definition, we can always work with the charts that "map out" the manifold so that the generalized coordinates are always elements of $\mathbb{R}^n$.] The *state* of a system at some time t is then described by n generalized coordinates, $q^k(t)$ and n generalized velocities, $v^k(t) = \dot{q}^k(t)$. In other words, the state of a system is described by a point $(q(t), v(t))$ in the space of all generalized coordinates and velocities called the *tangent bundle*, TC, of configuration space, where the coordinates $q(t)$ live as points on the configuration space and the velocities live as points on the tangent spaces to points on the configuration space. Equivalently, the time evolution of a classical system in the Lagrangian formulation can be described by a path $\gamma : \mathbb{R} \to C$ through configuration space. Lets call the path that is physically realized for a given system $\bar{\gamma}$ to distinguish it from some arbitrary path γ.

The physical essence of Lagrangian mechanics is that the dynamics of a system is determined by an action principle, which says that the time evolution of a classical system are those paths through configuration space which extremize a quantity with dimensions of $\hbar$ called the action

$$S[\gamma] = \int_{t_i}^{t_f} dt\; L(q(t), v(t), t), \tag{1}$$

where the simple combination of total kinetic and potential energies $L = T - V$ is called the Lagrangian of the system. If the Lagrangian of any system is known, then one can in principle solve for the dynamics of the system. The action principle — often called Hamilton's principle — is realized mathematically by finding, not the critical *points* as in ordinary calculus, but the critical *paths*, i.e., the actual paths that a system takes through configurations space which extremize the action. Extremizing an ordinary function $f(x)$ involves finding those points x where the first derivative $\frac{df}{dx}$ is zero and then making sure that that point is a global minimum or maximum. The action, however, is not a function of a number-valued variable $x \in \mathbb{R}^n$ but a *functional* of paths $\gamma : \mathbb{R} \to C$, i.e., a function of path-valued variables. Therefore, extremizing $S[\gamma]$ involves finding those paths $\bar{\gamma}$ for which the first variation δS — the functional analogue of the first derivative — is zero. Equivalently, the critical paths are obtained from demanding that the action is invariant to first order when paths are varied $\gamma \to \gamma + \epsilon\delta\gamma$. Referring to the Appendix **??** for more details, we can write the action of the varied paths as

$$S[\gamma + \epsilon\delta\gamma] = S[\bar{\gamma}] + \epsilon\delta S[\bar{\gamma}] + O(\epsilon^2), \tag{2}$$

where the first variation of the action, $\delta S[\bar{\gamma}]$ is

$$\begin{aligned} \delta S[\bar{\gamma}] &= \int_{t_i}^{t_f} dt\, \delta L(\gamma(t), \dot{\gamma}(t), t) \\ &= \int_{t_i}^{t_f} dt \left(\left.\frac{\partial L}{\partial q^k}\right|_{\bar{\gamma}} \delta q^k(t) + \left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} \delta \dot{q}^k(t) \right). \end{aligned} \tag{3}$$

We can integrate the second term by parts to obtain

$$\delta S[\bar{\gamma}] = \int_{t_i}^{t_f} dt \left(\left.\frac{\partial L}{\partial q^k}\right|_{\bar{\gamma}} - \frac{d}{dt} \left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} \right) \delta q^k(t) + \left(\left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} \delta q^k(t) \right) \Bigg|_{t_i}^{t_F}, \tag{4}$$

which simplifies to

$$\delta S[\bar{\gamma}] = \int_{t_i}^{t_f} dt \left(\left.\frac{\partial L}{\partial q^k}\right|_{\bar{\gamma}} - \frac{d}{dt} \left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} \right) \delta q^k(t) \tag{5}$$

since $\delta q^k(t)$ at the times $t = t_i$ and $t = t_f$ is zero. Therefore, since $\delta q^k(t)$ can be arbitrary for $t \in [t_i, t_f]$, the integrand of the first order term in the varied action must be zero. Therefore, finding the critical paths of the action amounts to finding the paths $\bar{\gamma}$, whose components $\bar{q}^k(t)$ satisfy the differential equation

$$\left.\frac{\partial L}{\partial q^k}\right|_{\bar{\gamma}} - \frac{d}{dt} \left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} = 0. \tag{6}$$

This differential equation is called the Euler-Lagrange equation. Once the Euler-Lagrange equation has been solved for the $\bar{q}^k(t)$'s, the cartesian coordinates of the particles in the system can then be recovered by the equations relating the generalized coordinates and the cartesian coordinates.

4 Hamiltonian

The third formulation of classical mechanics, the Hamiltonian formulation, introduces the notion of *phase space* Γ which is the space of all possible generalized coordinates and generalized momenta of a system. The equations of motion in this framework, called *Hamilton's equations* of motion, can be solved for q and p to obtain the trajectory of a system through phase space. These equations can be obtained by varying the action, but in terms of the coordinates and momenta. The *Hamiltonian* is the Legendre transform of the Lagrangian

$$H = \dot{q}^k p_k - L, \tag{7}$$

where everything is understood to be functions of the coordinates q, the generalized momenta $p = \frac{\partial L}{\partial \dot{q}}$, and possibly time t. Hamilton's equations are then

$$\begin{aligned} \dot{q}^k(t) &= \frac{\partial H}{\partial p_k} \\ \dot{p}_k(t) &= -\frac{\partial H}{\partial q^k}. \end{aligned} \tag{8}$$

Now, consider the time evolution of an observable (in classical mechanics, this is any smooth function of the phase space variables and possibly of time $O = O(q, p, t)$):

$$\frac{dO}{dt} = \frac{\partial O}{\partial q^k}\dot{q}^k + \frac{\partial O}{\partial p_k}\dot{p}_k + \frac{\partial O}{\partial t}.$$

Using Hamilton's equations for $\dot{q}$ and $\dot{p}$, we obtain

$$\frac{dO}{dt} = \frac{\partial O}{\partial q^k}\frac{\partial H}{\partial p_k} - \frac{\partial O}{\partial p_k}\frac{\partial H}{\partial q^k} + \frac{\partial O}{\partial t}.$$

If we define the Poisson bracket to be

$$\{A, B\} = \frac{\partial A}{\partial q^k}\frac{\partial B}{\partial p_k} - \frac{\partial A}{\partial p_k}\frac{\partial B}{\partial q^k},$$

the time evolution equation is

$$\frac{dO}{dt} = \{O, H\} + \frac{\partial O}{\partial t}.$$

The time evolution equation is particularly simple in form when the observable does not explicitly depend on time. Then, $\dot{O} = \{O, H\}$. In this sense, the Hamiltonian generates time evolution, i.e., if the Hamiltonian of a system is known, then the time evolution of the system is known.

In particular, the time evolution for q and p are of course Hamilton's equations, but they can be written as

$$\begin{aligned} \dot{q}^k(t) &= \{q^k, H\} \\ \dot{p}_k(t) &= \{p_k, H\}, \end{aligned} \tag{9}$$

which give Hamilton's equations in their original form as is easily verifiable.

A feature of the phase space variables (q, p) is that they are canonically conjugate, i.e.,

$$\{q^i, q^j\} = \{p_k, p_\ell\} \equiv 0 \quad \text{and} \quad \{q^k, p_\ell\} = \delta^k_\ell. \tag{10}$$

If a transformation of variables $(q, p) \to (Q, P)$ is made such that

$$\{Q^i, Q^j\} = \{P_k, P_\ell\} \equiv 0 \quad \text{and} \quad \{Q^k, P_\ell\} = \delta^k_\ell \tag{11}$$

then the transformation is called a canonical transformation, which preserves the dynamics of the system.

A Symplectic Formulation of Mechanics

A good review of the symplectic formulation of classical mechanics can be found in [1] or particularly the appropriate appendix in [2]. A great application of the symplectic structure to field theories, particularly general relativity, can be found in the introductory parts of [4]. Some knowledge of differential geometry is necessary to understand the symplectic formalism underlying classical mechanics.

There are many mathematical structures used in physics, such as the metric structure in general relativity of a Riemannian manifold, the structure of a connection on a fiber bundle in gauge theories in general, of the complex Kahler structure of a Calabi-Yau manifold in string theory. The mathematical structure underlying classical mechanics is called a symplectic structure.

The mathematical arena of classical mechanics is a symplectic manifold (Γ, Ω_{ab}), where Γ is an even-dimensional manifold called phase space, and the symplectic two-form Ω_{ab} gives the symplectic structure of the space, such that

- $d\Omega_{abc} = 0 \quad \text{or} \quad \nabla_{[a}\Omega_{bc]} = 0,$
- $\Omega_{ab}v^a = 0 \iff v^a = 0.$

For finite dimensional phase spaces, the second condition on Ω_{ab}, degeneracy, guarantees that there exists a unique inverse Ω^{ab}, such that $\Omega^{ab}\Omega_{bc} = \delta^a_c$. This means that there is an isomorphism between the tangent and cotangent bundles of phase space, so that the symplectic two-form or its inverse can be used to map vectors to co-vectors or co-vectors to vectors. That is, the symplectic structure can raise or lower indices of vectors and co-vectors. and we can use the symplectic two-form to raise and lower indices of tensors.

The existance of the symplectc structure on phases space allows for the specification of the time evolution of a system - paths in phase space - by functions on phase space. Since these functions provide the time evolution for a system, they are called the Hamiltonian functions, or simply Hamiltonians. These functions H construct the vectors fields whose integral curves are the time evolution of a system through phase space. This is done by using the derivative operator and symplectic structure on Γ,

$$X^a_H = \Omega^{ab}\nabla_b H.$$

Such vector fields are called a Hamiltonian vector field of H. Often, it is said that the symplectic structure is given by the Poisson brackets of a theory. This is evident from the definition of Poisson brackets in terms of the symplectic two-form

$$\{f, g\} = \Omega^{ab}\nabla_a f \nabla_b g.$$

This can also be written in terms of Lie derivative along Hamiltonian vector fields

$$\{f, g, \} = -\mathcal{L}_{X_f} g = \mathcal{L}_{X_g} f.$$

The dynamics of an observable on phase space, is also given by the symplectic structure,

$$\dot{f} = \mathcal{L}_{H_H} f = \{f, H\}.$$

The condition on Ω_{ab} that $d\Omega = 0$ allows Ω_{ab} to be written in terms of a symplectic potential ω_a: $\Omega_{ab} = 2\nabla_{[a}\omega_{b]}$. Thus, Ω_{ab} is determined up to the addition of a gradient.

Example:

As an example of this formalism, lets study how the usual expressions for things such as the symplectic potential, the symplectic two-form, and the equations of motion arise from this formalism. We consider the example where the state of a system can be represented by configurations and momentum variables (q^i, p_i), where q^i are coordinates on the configuration space C of a system and the momenta are the coordinates of (co-)tangent vectors based at q. These sort of choices of variables give the phase space a natural co-tangent bundle structure, $\Gamma = T^*C$. Since we can study any system using coordinate charts on phase space, this example is actually quite general.

The sympectic potential is then $\omega_a = p_i \nabla_a q^i$, from which we can derive the symplectic structure

$$\Omega_{ab} = 2\nabla_{[a} p_i \nabla_{b]} q^i.$$

What this formula for Ω_{ab} means in more familiiar terms is that, in the basis $(\nabla_a q, \nabla_a p)$ for the co-tangent bundle Γ, the symplectic two-form cna be written in matrix form as

$$[\Omega_{ij}] = \begin{bmatrix} 0 & -\mathbb{1} \\ \mathbb{1} & 0 \end{bmatrix}.$$

B Mathematics

Consider a function $f : \mathbb{R} \to \mathbb{R}$ at some point $x \in \mathbb{R}$, $f(x)$. Now, consider a variation of that point $x \to x + \epsilon\delta x$, so that the function at the varied point can be written in terms of the function and its derivatives at the original point and the infinitesimal variation parameter ϵ as a taylor series

$$f(x + \epsilon\delta x) = f(x) + \epsilon f'(x)\delta x + \frac{\epsilon^2}{2} f''(x)\delta x^2 + \frac{\epsilon^3}{3!} f^{(3)}(x)\delta x^3 + \cdots = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!} f^{(n)}(x)\delta x^n. \tag{12}$$

The first variation δf is defined to be the term that is first order in ϵ:

$$\delta f = \frac{d}{d\epsilon} f(x + \epsilon\delta x)\bigg|_{\epsilon=0} = f'(x)\delta x. \tag{13}$$

The second variation $\delta^2 f$ is defined to be the term which is second order in ϵ:

$$\delta^2 f = \frac{d^2}{d\epsilon^2} f(x + \epsilon\delta x)\bigg|_{\epsilon=0} = f''(x)\delta x^2.$$

In general, the n-th order variation $\delta^n f$ is defined to be the term which is n-th order in ϵ:

$$\delta^n f = \frac{d^n}{d\epsilon^n} f(x + \epsilon\delta x)\bigg|_{\epsilon=0} = f^{(n)}(x)\delta x^n.$$

If f is a function of multiple variables, $f(x_1, x_2, ..., x_n)$, then, when we vary $x^i \to x^i + \epsilon\delta x^i$, the function at the new point $\mathbf{x} + \epsilon\delta\mathbf{x}$ can be written as

$$f(\mathbf{x}+\epsilon\delta\mathbf{x}) = f(\mathbf{x}) + \epsilon\delta f(\mathbf{x}) + \frac{\epsilon^2}{2}\delta^2 f(\mathbf{x}) + \frac{\epsilon^3}{3!}\delta^3 f(\mathbf{x}) + O(\epsilon^4),$$

where

$$\delta f(\mathbf{x}) = \delta x^i \partial_i f(\mathbf{x}), \qquad \delta^2 f(\mathbf{x}) = \delta x^i \delta x^j \partial_i \partial_j f(\mathbf{x}), \qquad \delta^3 f(\mathbf{x}) = \delta x^i \delta x^j \delta x^k \partial_i \partial_j \partial_k f(\mathbf{x}), \qquad \text{etc.}$$

We can write this equation in matrix notation, using the Hessian matrix $\mathbf{H}_f$, as

$$f(\mathbf{x}+\epsilon\delta\mathbf{x}) = f(\mathbf{x}) + \epsilon\nabla f(\mathbf{x})\cdot\delta\mathbf{x} + \frac{\epsilon^2}{2}\delta\mathbf{x}^T\mathbf{H}_f\delta\mathbf{x} + O(\epsilon^3).$$

Now, consider a *functional* $S[x]$ which depends on a path, $x(t) \in \mathcal{M}$. Upon varying the path to $x(t) + \epsilon\delta(t)$, the action is

$$S[\gamma] = S[\bar{\gamma}] + \epsilon\delta S[\bar{\gamma}] + \frac{\epsilon^2}{2}\delta^2 S[\bar{\gamma}] + O(\epsilon^3).$$

If the functional S is of the integral form

$$S[\gamma] = \int dt\, L(\gamma(t), \dot{\gamma}(t), t),$$

then the first variation $\delta S[\bar{\gamma}]$ is

$$\delta S[\bar{\gamma}] = \int dt \left(\left.\frac{\partial L}{\partial q^k}\right|_{\bar{\gamma}} \delta\gamma^k(t) + \left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} \delta\dot{\gamma}^k(t) \right),$$

where by the evaluation $(\cdots)\Big|_{\bar{\gamma}}$ we really mean the evaluation of the functions at the critical path $\bar{\gamma}(t)$ and its velocity $\dot{\bar{\gamma}}(t)$. The requirement that $S[\bar{\gamma}]$ is invariant to first order,

$$S[\bar{\gamma} + \epsilon\delta\gamma] = S[\bar{\gamma}] + O(\epsilon^2),$$

is the condition for an extremum of the functional S. Since the function $\delta\gamma(t)$ can be arbitrary, the requirement that $\delta S[\bar{\gamma}]$ vanishes is equivalent to the differential equation

$$\left.\frac{\partial L}{\partial q^k}\right|_{\bar{\gamma}} - \frac{d}{dt}\left.\frac{\partial L}{\partial \dot{q}^k}\right|_{\bar{\gamma}} = 0.$$

This equation is usually called the Euler-Lagrange equation in the context of classical mechanics. If the integrand of $S[\gamma]$ depends on higher order derivatives — i.e., $L = L(\gamma(t), \gamma^{(1)}(t), ..., \gamma^{(N)})$ — and the variations $\delta\gamma^{(\alpha)}(t)$ vanish on the boundary of the paths for $\alpha = 0, 1, ..., N-1$, then the Euler-Lagrange equations generalize to

$$\sum_{\alpha=0}^{N} (-1)^\alpha \frac{d^\alpha}{dt^\alpha}\left(\frac{\partial L}{\partial q^{(\alpha)\,k}}\right) = 0. \tag{14}$$

This generalized equation is derived by considering the first variation:

$$\delta S[\bar{\gamma}] = \int_{t_i}^{t_f} dt \left(\frac{\partial L}{\partial q^k}\bigg|_{\bar{\gamma}} \delta\gamma^k(t) + \cdots + \frac{\partial L}{\partial q^{(\alpha)k}}\bigg|_{\bar{\gamma}} \delta\gamma^{(\alpha)k}(t) + \cdots + \frac{\partial L}{\partial q^{(N)k}}\bigg|_{\bar{\gamma}} \delta\gamma^{(N)k}(t) \right)$$
$$- \int_{t_i}^{t_f} dt \left(\sum_{\alpha=0}^{N} (-1)^\alpha \frac{d^\alpha}{dt^\alpha} \frac{\partial L}{\partial q^{(\alpha)k}}\bigg|_{\bar{\gamma}} \right) \delta\gamma^k(t) + \sum_{\alpha=0}^{N} \sum_{\beta=0}^{\alpha-1} \left((-1)^\beta \frac{d^\beta}{dt^\beta} \frac{\partial L}{\partial q^{(\alpha)k}}\bigg|_{\bar{\gamma}} \delta\gamma^{(\alpha-\beta-1)k}(t) \right) \Bigg|_{t_i}^{t_f}.$$

If we demand that the variations $\delta\gamma^k(t)$, $\delta\gamma^{(1)k}(t)$, ..., $\delta\gamma^{(N-1)k}(t)$ are zero at the boundary of the paths (i.e., at time $t = t_i, t_f$), then the requirement that the first variation of S vanishes is a generalized Euler-Lagrange equation for each degree of freedom $k = 1, 2, ..., n$:

$$\sum_{\alpha=0}^{N} (-1)^\alpha \frac{d^\alpha}{dt^\alpha} \frac{\partial L}{\partial q^{(\alpha)k}}\bigg|_{\bar{\gamma}} = 0.$$

This equation is not merely of mathematical importance: it is actually useful in physics. It is relevant when a Lagrangian depends on second and higher derivatives. Examples of such higher-order Lagrangians are the Lagrangian for an elastic rod, the Lagrangian for gravity (the Einstein-Hilbert Lagrangian), and most Lagrangians for systems with stresss and strain.

C Generalized Lagrange Equations

We will derive the generalized Euler-Lagrange equations for a "Lagrangian" $L = L(x(t), x'(t), ..., x^{(n)}(t))$, and associated "action" functional

$$S[x] = \int_{t_i}^{t_f} dt L(x(t), x^{(1)}(t), ..., x^{(n)}(t)).$$

Upon varying the path $x \to x + \epsilon\delta x$, the action becomes

$$\delta S[x] = \int dt \left[\frac{\partial L}{\partial x}\delta x + \frac{\partial L}{\partial x^{(1)}}\delta x^{(1)} + \cdots + \frac{\partial L}{\partial x^{(k)}}\delta x^{(k)} + \cdots + \frac{\partial L}{\partial x^{(n)}}\delta x^{(n)} \right].$$

Lets focus on the k-th term $\int dt \frac{\partial L}{\partial x^{(k)}}\delta x^{(k)}$:

$$\begin{aligned}
\int dt \frac{\partial L}{\partial x^{(k)}}\delta x^{(k)} &= -\int dt \frac{d}{dt}\frac{\partial L}{\partial x^{(k)}}\delta x^{(k-1)} + \left(\frac{\partial L}{\partial x^{(k)}}\delta x^{(k-1)} \right) \Bigg|_{t_i}^{t_f} \\
&= \int dt \frac{d^2}{dt^2}\frac{\partial L}{\partial x^{(k)}}\delta x^{(k-2)} - \left(\frac{d}{dt}\frac{\partial L}{\partial x^{(k)}}\delta x^{(k-2)} \right) \Bigg|_{t_i}^{t_f} + \left(\frac{\partial L}{\partial x^{(k)}}\delta x^{(k-1)} \right) \Bigg|_{t_i}^{t_f} \\
&\vdots \\
&= (-1)^k \int dt \frac{d^k}{dt^k} \left(\frac{\partial L}{\partial x^{(k)}} \right) \delta x + (-1)^{k-1} \left(\frac{d^{k-1}}{dt^{k-1}} \left(\frac{\partial L}{\partial x^{(1)}} \right) \delta x \right) \Bigg|_{t_i}^{t_f} \\
&\quad + (-1)^{k-2} \left(\frac{d^{k-2}}{dt^{k-2}} \left(\frac{\partial L}{\partial x} \right) \delta x^{(1)} \right) \Bigg|_{t_i}^{t_f} + \cdots + \left(\frac{\partial L}{\partial x^{(k)}}\delta x^{(k-1)} \right) \Bigg|_{t_i}^{t_f} \\
&= (-1)^k \int_{t_i}^{t_f} dt \frac{d^k}{dt^k} \left(\frac{\partial L}{\partial x^{(k)}} \right) \delta x + \sum_{\ell=1}^{k} (-1)^{k-\ell} \frac{d^{k-\ell}}{dt^{k-\ell}} \left(\frac{\partial L}{\partial x^{(\ell)}}\delta x^{(\ell-1)} \right) \Bigg|_{t_i}^{t_f}.
\end{aligned}$$

Therefore, we have that

$$\delta S[x] = \int_{t_i}^{t_f} dt \sum_{k=0}^{n} (-1)^k \frac{d^k}{dt^k}\left(\frac{\partial L}{\partial x^{(k)}}\right)\delta x + \sum_{k=1}^{n}\sum_{\ell=1}^{k}(-1)^{k-\ell}\frac{d^{k-\ell}}{dt^{k-\ell}}\left(\frac{\partial L}{\partial x^{(\ell)}}\right)\delta x^{(\ell-1)}\Bigg|_{t_i}^{t_f}. \tag{15}$$

If we demand that the variations at the boundary is zero, we then have the generalized Euler-Lagrange equations

$$\sum_{k=0}^{n}(-1)^k \frac{d^k}{dt^k}\left(\frac{\partial L}{\partial x^{(k)}}\right) = 0. \tag{16}$$

If we do not impose some of the $4n$ boundary conditions, the equations will provide the boundary conditions for us. Such boundary conditions are called the *free* boundary conditions. They, in general, give different solutions for the critical path(s) $x(t)$ than the Dirichlet boundary conditions.

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